

Horizon Forecast

Driver-First Cascaded Nowcasting of Clouds & Rain over Israel

Or Mordechay Hod · Gilad Boudman

Advisors: Mrs. Elena Kramer, Dr. Dan Lemberg · Project 26-1-R-1 · Capstone Project Phase B



INTRODUCTION

Horizon Forecast is a deep-learning **nowcasting** system that predicts the near-future evolution of wind, temperature, cloud structure and rain over Israel from recent weather-satellite imagery.

Its novelty - the **Driver-First Cascade** - predicts the physical drivers (wind, temperature) first, then conditions the cloud and rain forecast on them, enforcing physical causality in the architecture rather than only in the loss.

REQUIREMENTS

Functional

- 3-layer forecast map - drivers (wind+temp), cloud structure, rain intensity
- Multi-channel input - satellite IR/WV + DEM
- Dual-source validation - future satellite + IMS stations
- Chronological splits (no leakage), report RMSE + CSI

Non-functional

- Real-time inference - faster than the 15 - min data interval
- Consumer-grade GPU - fits desktop VRAM, no data-center hardware
- Efficient training - GPU saturated, I/O never starves it
- Reproducible runs - deterministic seeds, fixed splits

RESEARCH QUESTIONS & ANSWERS

RQ1. Does predicting physical drivers (wind + temp) first improve rain nowcasting vs an end-to-end model?
→ **YES** - Driver-First beats End-to-End by +52% rain-CSI on unseen data, at every horizon (T+15...T+120).

RQ2. Does the Driver-First Cascade improve cloud-structure forecasting vs an end-to-end model?
→ **NO** - both tie at SSIM $\approx$ 0.80. Clouds are already well-recovered straight from the satellite frames, so the driver path leaves them unchanged.

Physics pays off only for rain, where it's needed most.

DATA & ARCHITECTURE

Data:

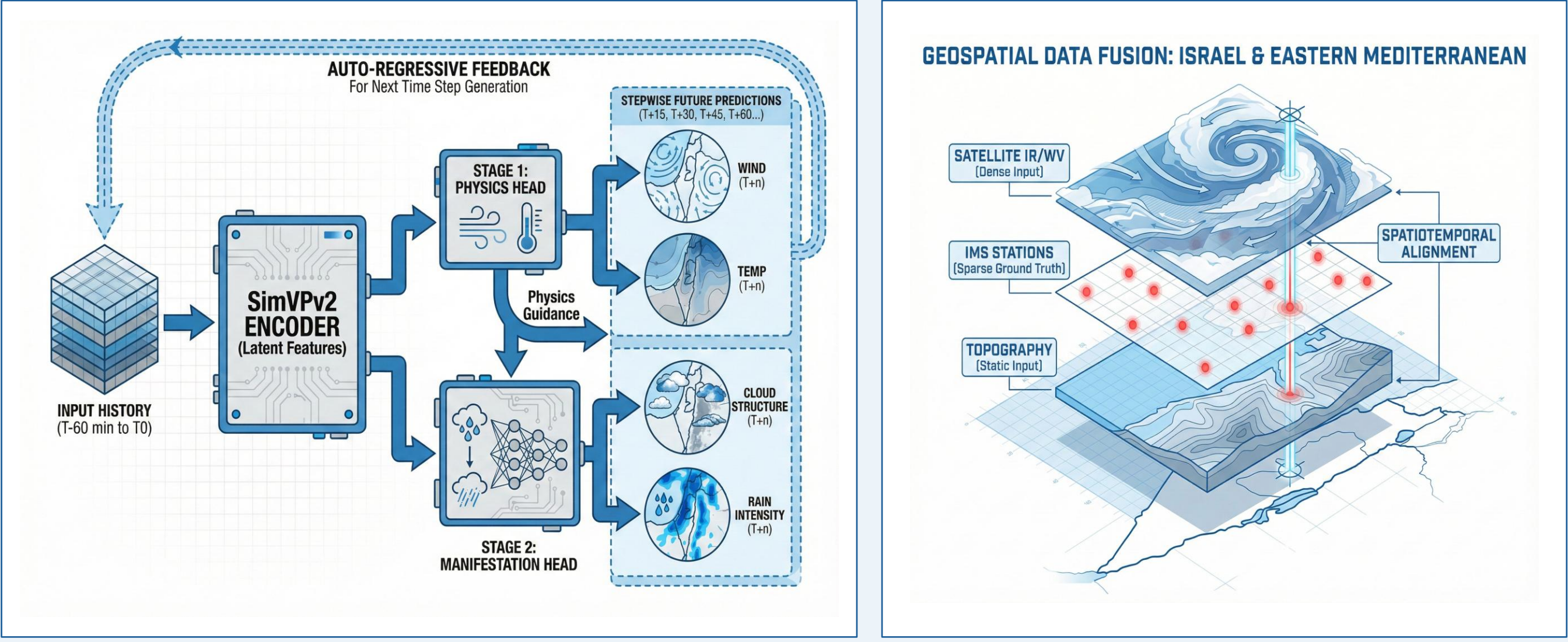
- Inputs:** SEVIRI satellite IR + water-vapor (15-min) and SRTM terrain (DEM). Targets ERA5 dense wind/temp fields and IMS rain gauges (~100 stations).
- Splits:** train 2020 - 2023, validate Jan - Jun 2024, test Jul 2024 → 2025 (chronological).

Hybrid Cascade:

SimVPv2 gSTA encoder → Stage-1 Physics Driver Head (wind/temp) → Stage-2 Manifestation Head: cloud head + transformer patch - attention rain head (64 buckets).

Technologies & tools:

PyTorch (BF16, grad-checkpointing), NumPy/SciPy/pandas/scikit-image/OpenCV, Matplotlib, ERA5+SEVIRI+IMS+SRTM data, Tkinter + PyInstaller demo.



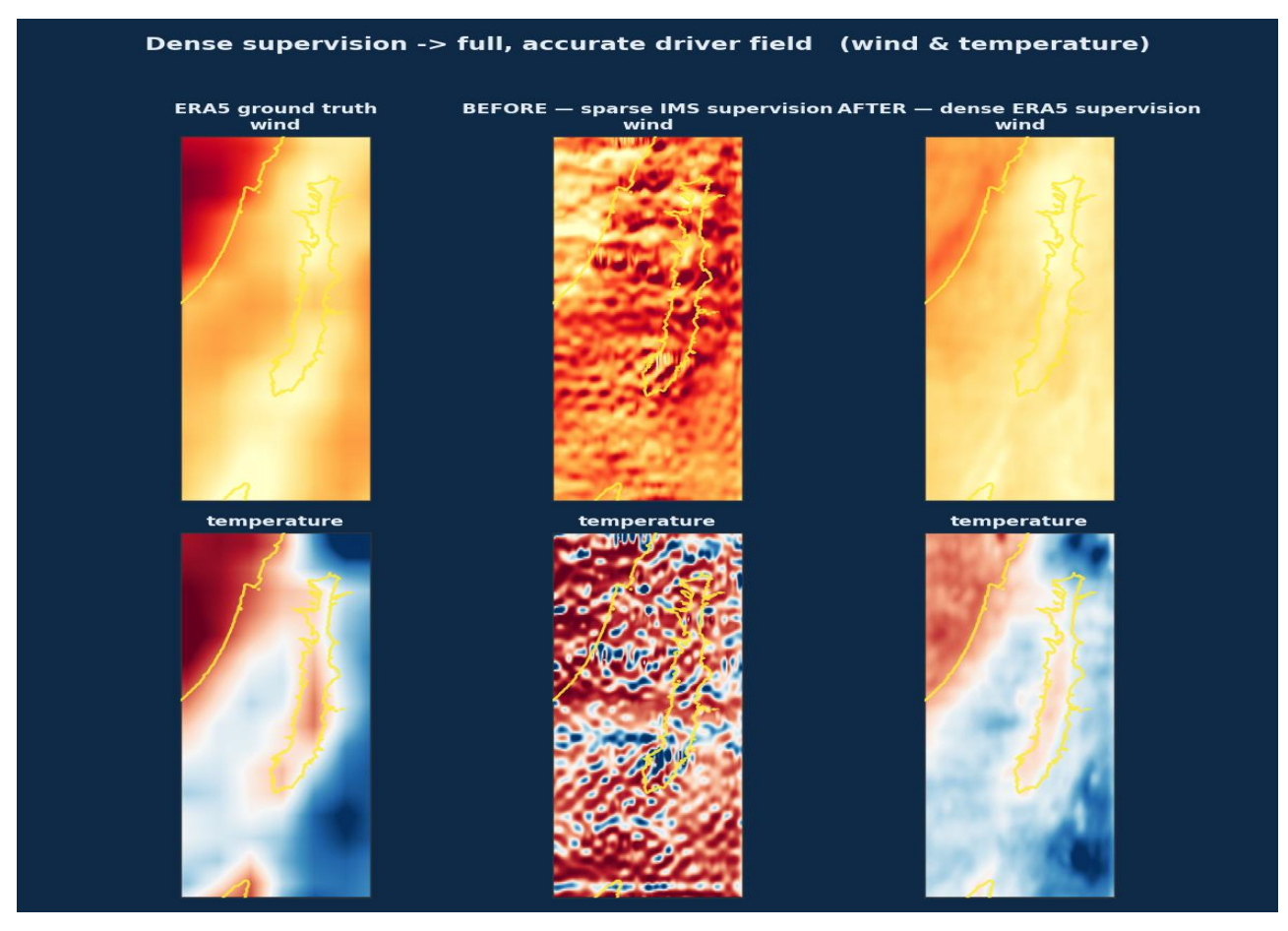
EXPERIMENTAL DESIGN

Hypothesis-driven experimentation (H1-H5):

- H1 - H3:** Tuned the driver loss-weights three ways - stronger, weaker, then split wind/temp - all failed under sparse supervision. Conclusion: the limit was the data, not the weights.
- H4:** Changed to ERA5 dense supervision from scattered IMS stations for phase 1.
- H5:** Two-phase frozen training (the final method, +52% CSI).

Dense ERA5 supervision:

(ECMWF's global reanalysis - a gridded, full-field weather record) replaces the ~100 scattered IMS stations as the wind/temp target. The dense grid made the predicted fields full and accurate, cutting driver error ~30×.



Two-phase frozen training:

- Phase 1:** Train the encoder + physics-driver head (cloud/rain head frozen).
- Phase 2:** Train the cloud + rain head (encoder/physics-driver head frozen).
- Separating the stages removes cross-stage gradient interference.**

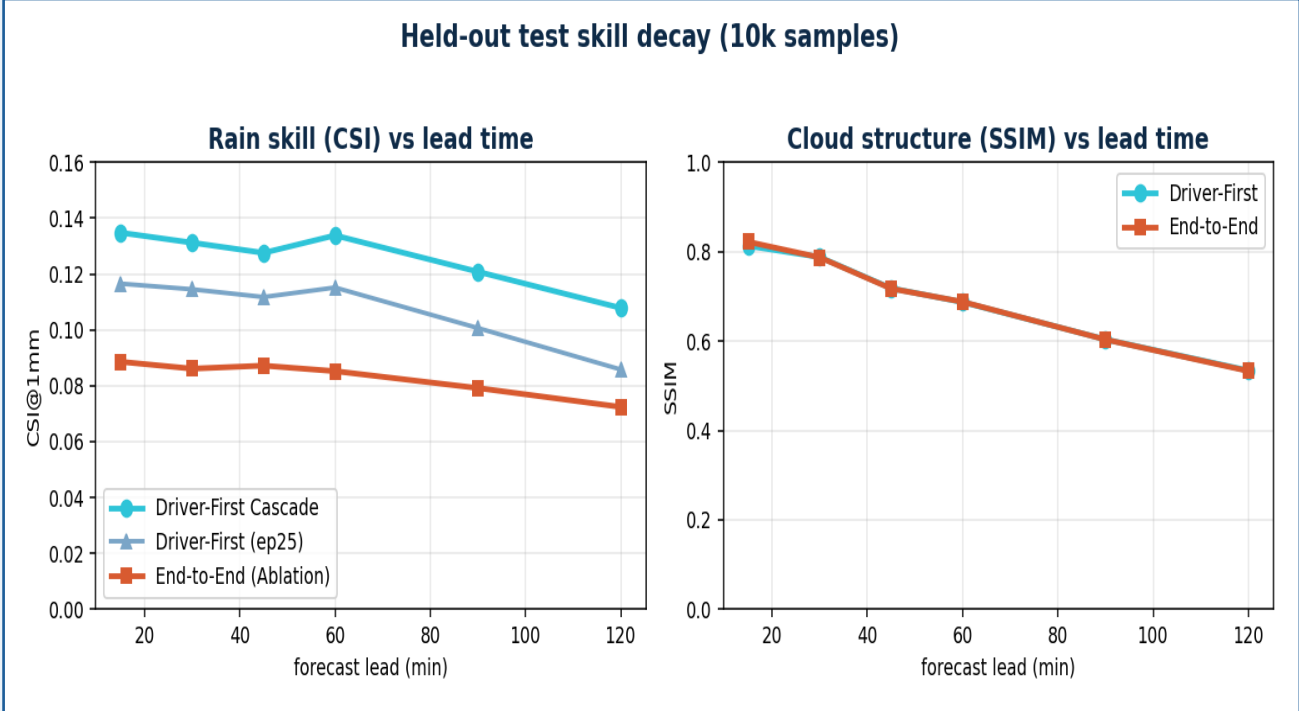
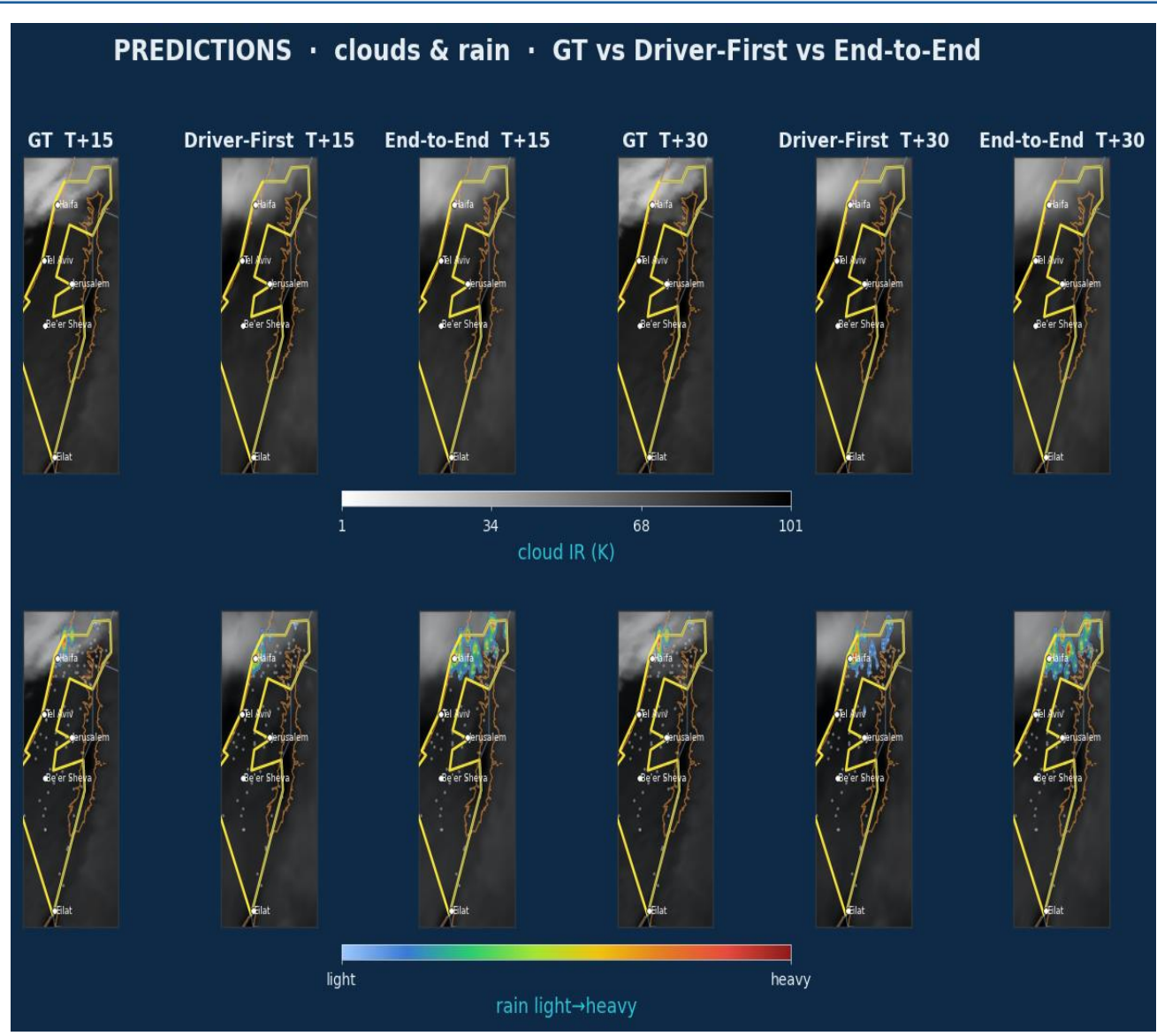
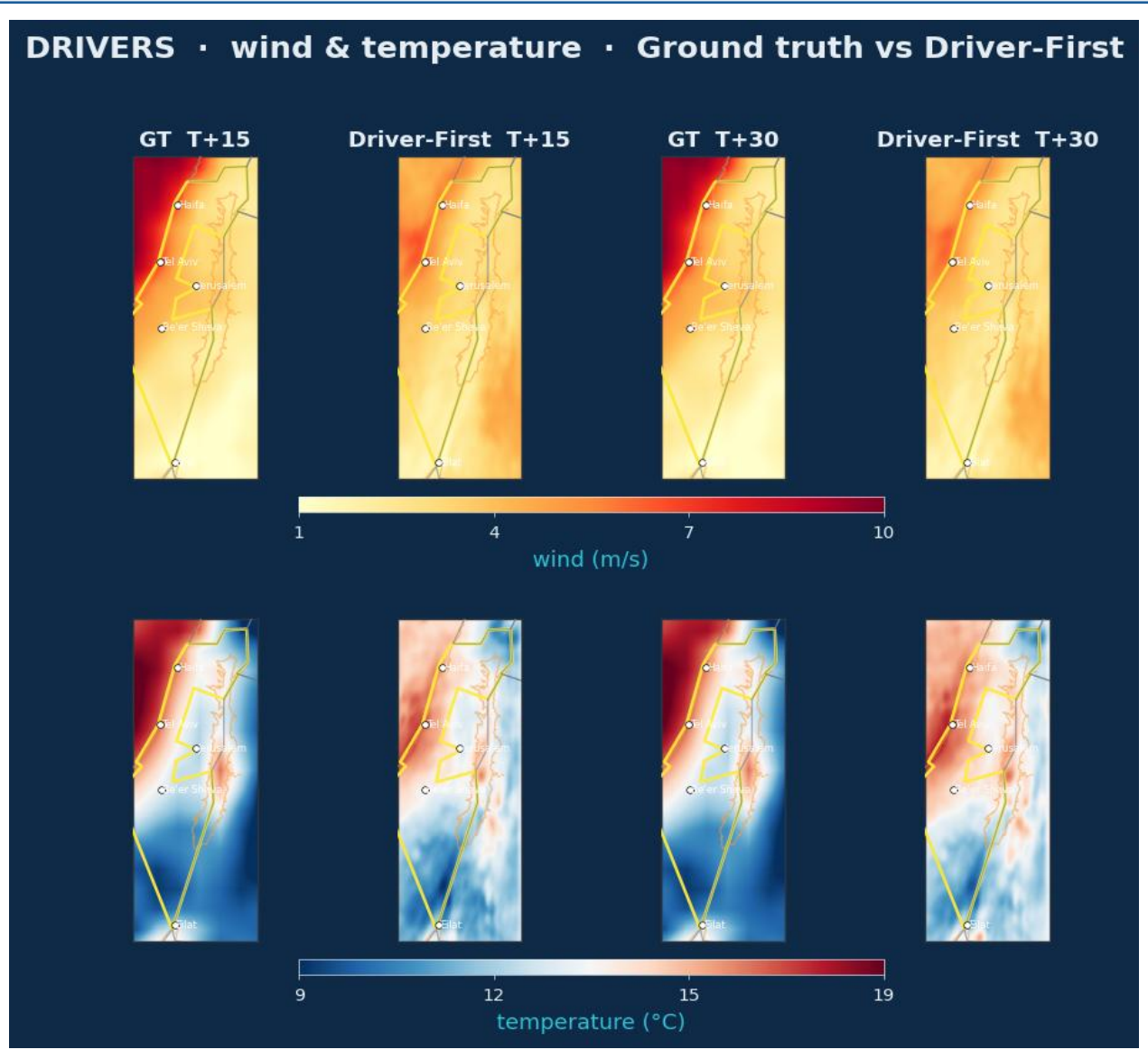
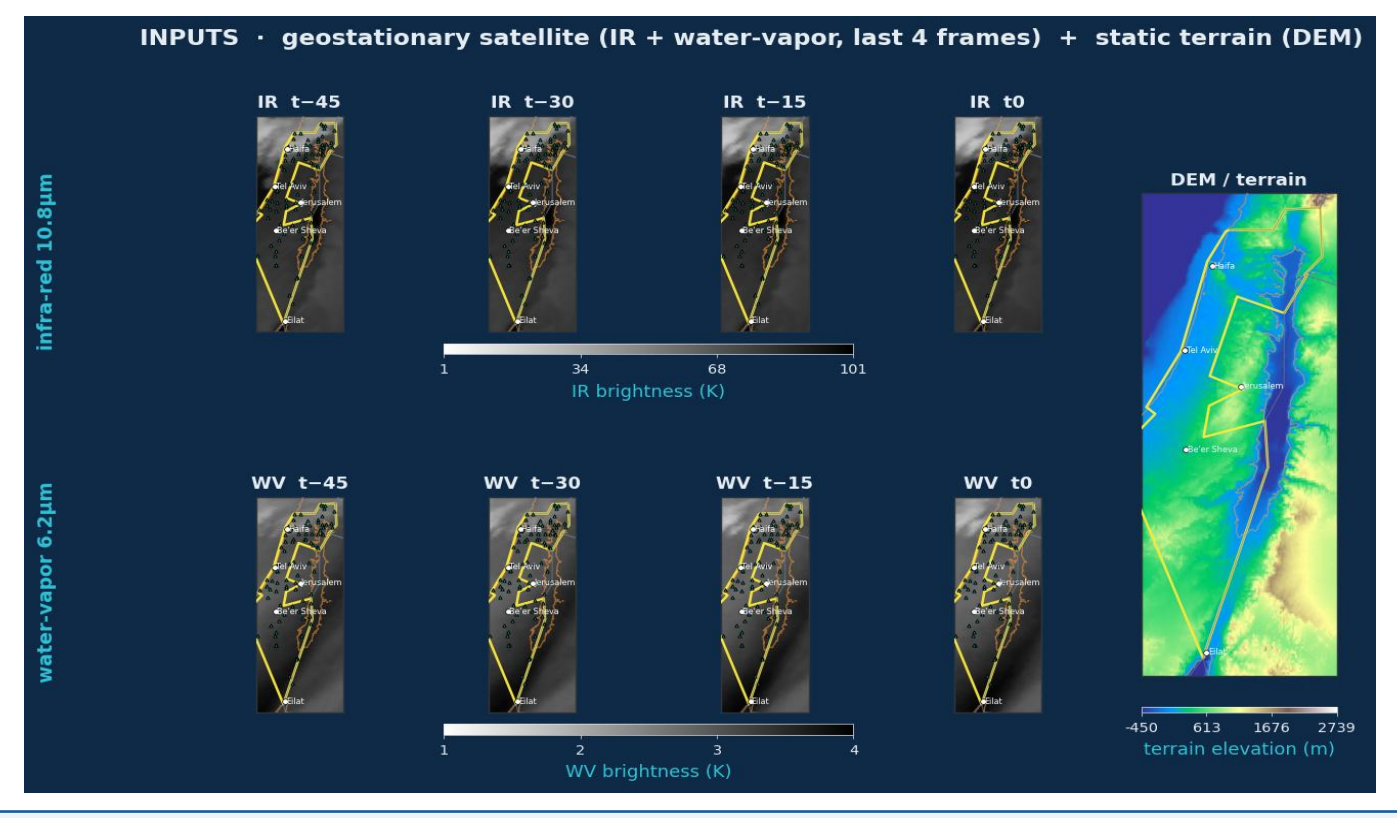
Fair ablation & evaluation:

Both models share the same architecture, training data, and hardware - in the End-to-End model, the rain head never receives the predicted drivers. With everything else held fixed. Evaluated on 52,566 forecasts the model never saw (Jul 2024→2025), measured by CSI, SSIM, RMSE, POD/FAR and rolled forward step-by-step to T+120.

RESULTS & FINDINGS

Forecast flow

- ① inputs →
- ② drivers →
- ③ predictions



Held-out test values: T+15 / T+30			
Model	CSI↑	SSIM=	FAR↓
Driver-First	.135/.131	.81/.79	.86/.86
End-to-End	.089/.086	.82/.79	.91/.91

+52%

rain-CSI vs ablation

0.135

Held-out CSI (vs 0.089)

CHALLENGES & CONSTRAINTS - (Data / Analytical)

IR/WV channel swap bug in the source data:

Every other frame had its infra-red and water-vapor channels silently swapped - a corruption inherited from the raw archive. We traced the source, repaired the channel alignment, and retrained all models on the fixed data.

Sparse rain ground truth caps absolute CSI :

Only ~100 IMS gauges measure rain, so the scoring target is thin - ERA5's dense grid rescued the drivers (wind/temp), but no equivalent dense truth exists for rain.

Extreme class imbalance:

~98.8% of station readings are dry - rain is ~1% and heavy rain a fraction of that. The model must over-predict to catch rare events, inflating false alarms (FAR 0.86-0.91) and dragging absolute CSI.

CHALLENGES & CONSTRAINTS - II (Engineering)

GPU & CPU-RAM OOM:

fixed with gradient checkpointing + a leaner data loader. Trained on RunPod (cloud) H100 + local RTX A5000.

Cross-stage gradient interference:

the rain and driver losses worked against each other under joint training. Splitting into two frozen phases fixed it.

FINDINGS & KEY LEARNINGS

Driver-first works - once the stages are separated. Predicting wind/temp before rain improves nowcasting skill, but only after two-phase training removes the conflict between the phases.

Select on CSI, not validation loss - lower loss didn't mean better forecasts. A controlled, matched ablation is essential to attribute the gain.

The cascade idea is correct and working - denser data is what's needed to push rain prediction further. Driver-first already proved itself by beating the end-to-end model on rain (+52% CSI), going higher now needs a denser rain reference, not a new architecture.

FUTURE WORK

Test the dense-rain-data theory - train on full-grid radar truth to confirm it raises absolute CSI.

Train on longer horizons - current training rolls out only to T+30, extending past it should improve the far horizon forecasts.

Adaptive weighting - let the model learn how to weight its losses/inputs instead of fixed weights.

Deploy the system - package the model for real-time operational use.